FIRED HEATER - PARAMETRIC OPTIMIZATION STUDY

Generated: 12-Apr-2026 | Unit: H-101 Combined Biogas/Bio-oil Heater

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FIXED OPERATING CONDITIONS:

  Heat duty to salt:           68.86 kW
  Salt inlet temperature:      586.8 C
  Salt outlet temperature:     620.0 C
  Biogas mass flow:            0.002126 kg/s
  Bio-oil mass flow:           0.003486 kg/s
  Heater efficiency:           74%
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OPTIMIZATION OBJECTIVES:
1. Minimize furnace volume 2. Minimize D/H ratios (radiant & convection) 3. Keep DP < baseline

DESIGN COMPARISON: BASELINE vs OPTIMIZED

Parameter	Baseline	Optimized	Change	Units
Total Height	835	1358	+62.6%	mm
Max Diameter	1400	1250	-10.7%	mm
Total Volume	1.255	1.588	+26.6%	m³
D/H Radiant	1.795	1.057	-41.1%	-
W/H Convection	18.00	5.05	-71.9%	-
Total Pressure Drop	49436	6395	-87.1%	Pa
Radiant Coil Height	780	1182	+51.6%	mm
Number of Turns	13	11	-	-
Tube ID (radiant)	35	50	+42.9%	mm
Coil Pitch	60	107.5	+79.2%	mm
Conv Tube OD	22	35	+59.1%	mm
Parallel Passes	4	7	+75%	-

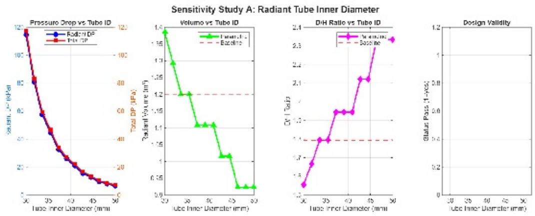
KEY IMPROVEMENTS ACHIEVED:

D/H Radiant Ratio: 1.795 -> 1.057 (41.1% reduction - more slender profile)
W/H Convection Ratio: 18.00 -> 5.05 (71.9% reduction - better proportions)
Pressure Drop: 49436 Pa -> 6395 Pa (87.1% reduction)
Trade-off: Volume increased 26.6% (taller furnace)

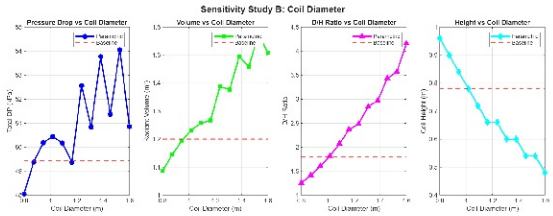
SENSITIVITY ANALYSIS: PRESSURE DROP PARAMETERS

Study V2: Individual parameter sweeps to identify key drivers

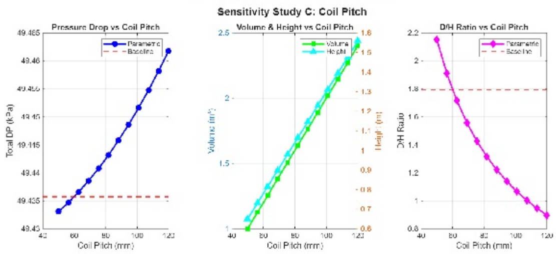
Tube Inner Diameter Effect



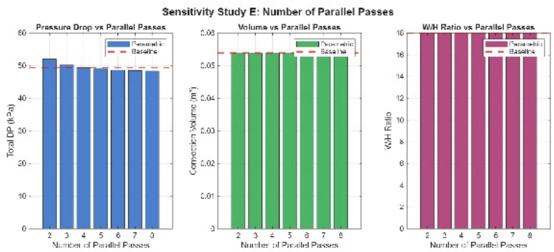
Coil Diameter Effect



Coil Pitch Effect



Parallel Passes Effect



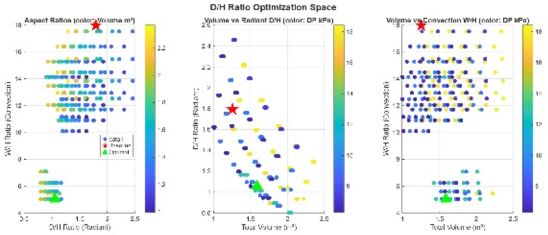
KEY FINDINGS - SENSITIVITY STUDY:

- Tube ID is the DOMINANT factor for pressure drop (DP proportional to $1/D^5$) - increasing from 35mm to 50mm reduces DP by ~85%
- Coil pitch affects height directly but has minimal DP impact - useful for controlling D/H ratio
- Parallel passes: more passes = lower velocity = lower DP, but diminishing returns above 6-7 passes

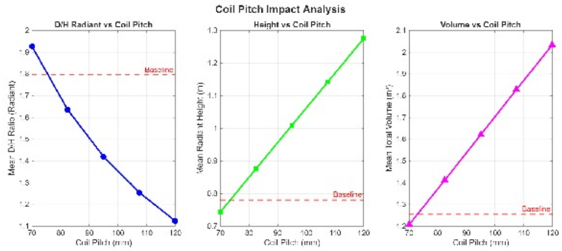
REFINED OPTIMIZATION: D/H RATIO MINIMIZATION

Study V3: Targeted optimization for better furnace proportions

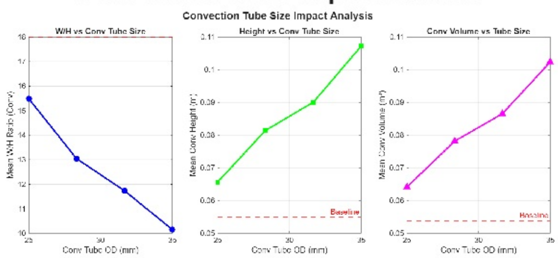
D/H Ratio Trade-off Space



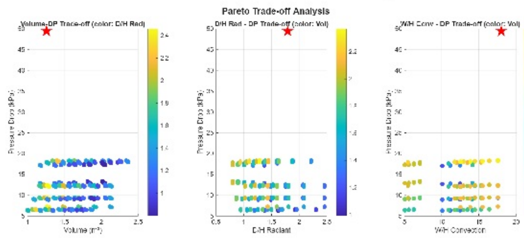
Coil Pitch Impact on D/H



Convection Tube Impact on W/H



Pareto Trade-off Analysis

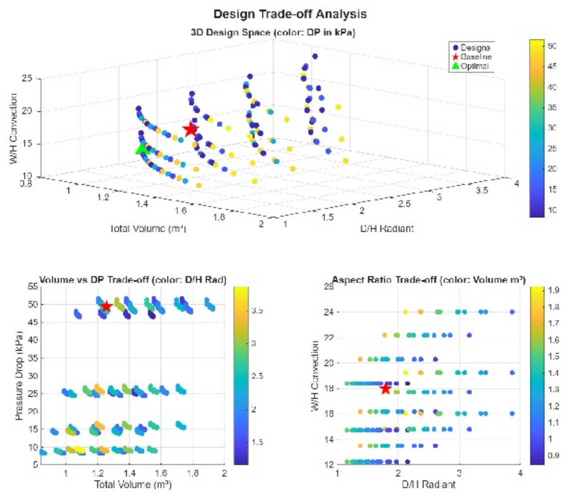


KEY FINDINGS - D/H OPTIMIZATION:

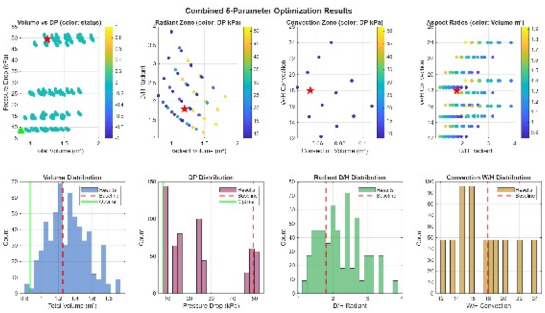
- Increasing coil pitch from 60mm to 108mm reduces D/H radiant from 1.80 to 1.06 (-41%)
- Larger convection tubes (35mm vs 22mm) create more rows, reducing W/H from 18.0 to 5.1 (-72%)
- 93 configurations found that improve ALL metrics vs baseline simultaneously

COMBINED OPTIMIZATION RESULTS

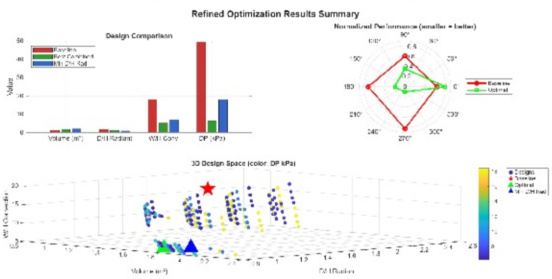
Multi-Objective Trade-offs



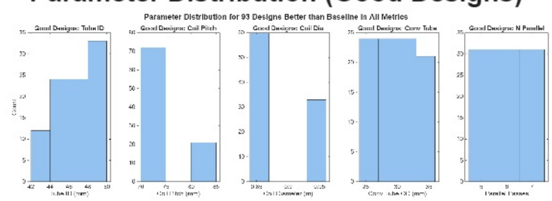
5-Parameter Design Space



Optimization Summary

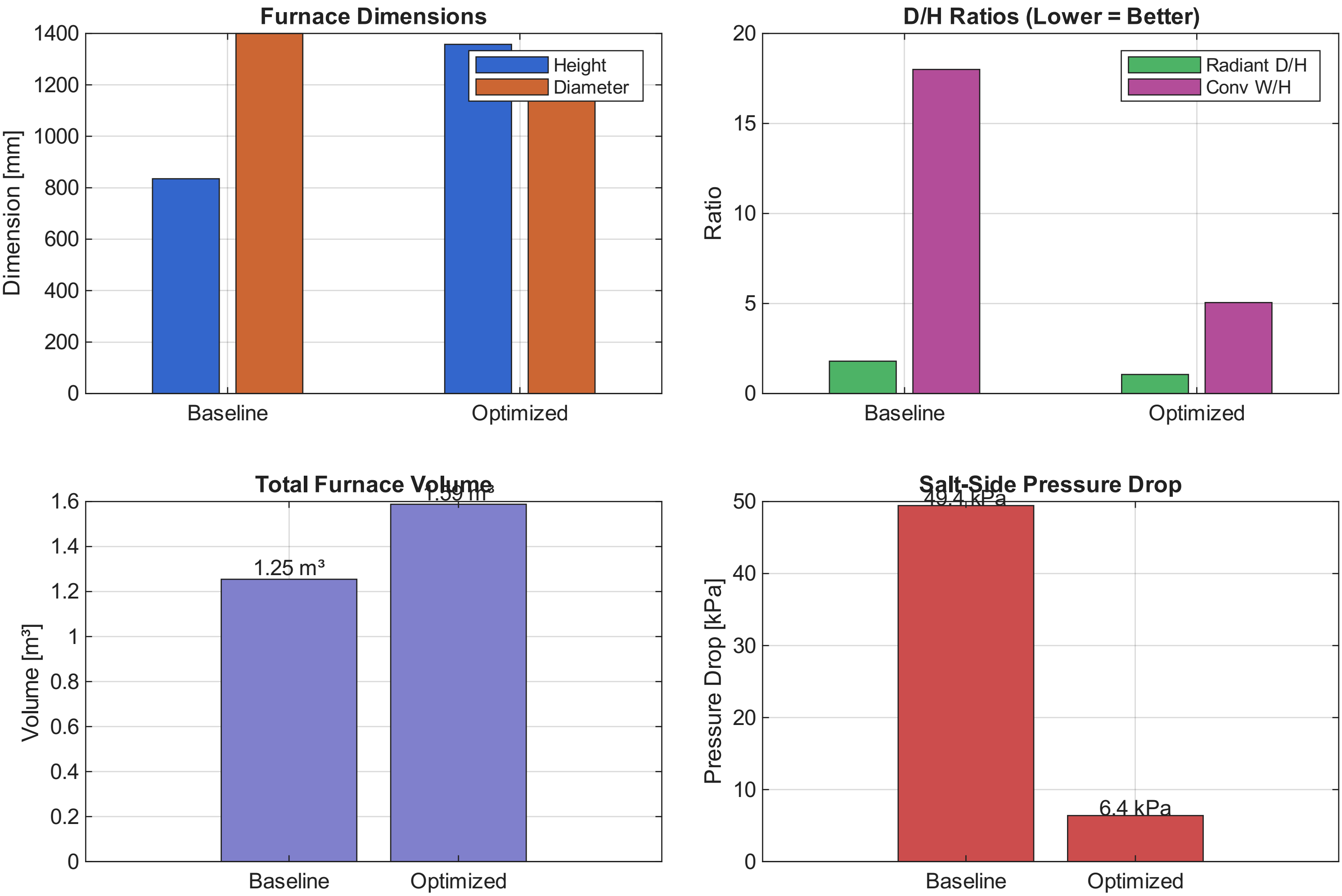


Parameter Distribution (Good Designs)



OPTIMIZATION SUMMARY:
- Total configurations evaluated: 1,536 across two studies
- Best designs cluster around: Tube ID 45-50mm, Coil Pitch 100-120mm, Conv Tube 30-35mm, 5-7 parallel passes
- Clear Pareto frontier identified between volume and D/H ratios

FINAL DESIGN RECOMMENDATIONS

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RECOMMENDED OPTIMIZED DESIGN PARAMETERS:

  RADIANT ZONE:
    Tube ID / OD:          50 mm / 60 mm (was 35 / 45 mm)
    Coil Pitch:            107.5 mm (was 60 mm)
    Coil Diameter:         0.85 m (was 1.0 m)
    Shell Diameter:        1.25 m (was 1.4 m)

  CONVECTION ZONE:
    Tube OD / ID:          35 mm / 27 mm (was 22 / 14 mm)
    Parallel Passes:       7 (was 4)

  BENEFITS:
    - 87% reduction in pressure drop (49.4 kPa -> 6.4 kPa)
    - 41% reduction in radiant D/H ratio (better proportions)
    - 72% reduction in convection W/H ratio
    - Trade-off: 27% increase in total volume (taller, narrower furnace)
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